PASS 5 – NASA Hump CFD Optimization

Convergence-driven improvement with official experimental C_p/C_f comparison

Codex refinement pass in the local repository workspace

April 4, 2026

1 Objective

Pass five builds directly on the promoted pass-four workflow and asks a narrower question than the earlier passes: can the current best honest NASA hump reconstruction be improved further without changing the no-cheat rules, while keeping the official experimental C_p and C_f comparisons central to the diagnosis?

The pass-four best benchmark MAE was `0.1659136747`. Pass five targeted three physically motivated paths:

- push the best SST baseline much farther in convergence before changing physics;
- test one official-grid-inspired geometry and top-wall refinement;
- screen one additional turbulence model that is actually available in the local Dockerized OpenFOAM installation.

2 What Pass Four Taught Us

Pass four established four important facts that shaped the pass-five plan:

- the direct machine-readable NASA C_p and C_f tables can be integrated honestly without tracing plots or interpolating target curves;
- the analytic turbulent-boundary-layer inlet was physically reasonable, but it did not beat the simpler SST continuation;
- the front-half suction region and wall-shear recovery still mismatched the official experiment;
- deeper convergence of the best SST case mattered more than adding complexity.

Those points led to a conservative pass-five ranking:

1. continue the existing best SST case from `650` to later checkpoints;
2. improve geometry and mesh fidelity using the official no-plenum grid guidance;
3. screen `SpalartAllmaras` only if it can be run stably in the local OpenFOAM installation.

3 Audit Of The Current Workflow

The main pass-five control files remained:

- geometry and mesh generation: `scripts/generate_nasa_hump_blockmesh.py`
- case numerics, model selection, and convergence controls: `scripts/configure_nasa_hump_case.py`
- inlet field generation: `scripts/write_nasa_hump_inlet_profile.py`
- model-specific field preparation: `scripts/prepare_nasa_hump_model_fields.py`
- benchmark MAE evaluation: `scripts/evaluate_nasa_hump_case.py`
- wall C_p/C_f evaluation: `scripts/evaluate_nasa_hump_wall_metrics.py`
- pass-five experiment orchestration: `scripts/run_nasa_hump_pass5_experiments.py`
- pass-five figure generation: `scripts/make_nasa_hump_pass5_figures.py`
- ParaView rendering from the live `.foam` case: `scripts/render_nasa_hump_paraview.sh`

Even after pass four, three major approximations still mattered:

- the hump and upper-wall contour were still reconstructed rather than imported from an official no-plenum grid file;
- the promoted SST solution still improved when iterated longer, so convergence error had not been exhausted;
- the turbulence-model screen was limited by what the local OpenFOAM installation actually supports.

4 Methodology

Pass five stayed inside the approved source boundary:

- local repository contents;
- the approved NASA hump validation page;
- the approved NASA hump grids page;
- the approved NASA CFDVAL2004 case-3 experimental-data page;
- Dockerized OpenFOAM and locally authored scripts.

No hidden solution data were imported. No plot tracing, image digitization, NASA-curve interpolation, or post-hoc warping of CFD outputs was used. The official experimental wall data continued to come from the machine-readable tables already normalized under `data/experimental/NASA_hump`. The wall comparisons still use the strict nearest-sample rule rather than an interpolated CFD wall curve.

5 Geometry And Grid-Fidelity Improvements

The geometry-focused experiment did not replace the main workflow with an external grid import. Instead, it used the official no-plenum grid guidance to make the local structured reconstruction more faithful while preserving the repository's shell-driven OpenFOAM workflow.

The targeted geometry and mesh changes were:

- broaden the contoured upper-wall blockage region from a mild local dip to a wider streamwise influence;
- redistribute the streamwise splits so more cells fall into the front-half suction region, separation onset, and recovery region;
- increase the wall-normal resolution from the older pass-four level to a tighter targeted near-wall arrangement on the geometry-fidelity case.

The official-grid-inspired experiment used:

- `NASA_HUMP_X_SPLITS=-1.35,-0.58,-0.10,0.08,0.22,0.42,0.72,0.84`
- `NASA_HUMP_X_CELLS=110,120,110,140,160,150,30`
- `NASA_HUMP_Y_CELLS=240`
- `NASA_HUMP_TOP_CONTOUR_START=-0.58`
- `NASA_HUMP_TOP_CONTOUR_END=0.82`
- `NASA_HUMP_TOP_CONTOUR_DIP=0.03`

This made the top-wall contour and streamwise grid placement more faithful than the pass-four approximation, but the final metrics show that this particular refinement did not beat the already-promoted SST continuation.

6 Convergence Study

The most important pass-five experiment was a disciplined continuation of the promoted pass-four SST winner. Nothing physical was changed in this case. The fields, mesh, geometry, and turbulence closure were preserved and the solution was simply continued from `650` to `1200` iterations with evaluation checkpoints at `650`, `800`, `1000`, and `1200`.

Checkpoint	Benchmark MAE	C_p MAE	C_f MAE	Interpretation
650	0.1659136747	0.1711800326	0.0008476129	Pass-four winner carried into pass five.
800	0.1547381028	0.1641787966	0.0007681825	Strong improvement with no physics changes.
1000	0.1433377644	0.1578824614	0.0006999053	Continued simultaneous improvement in benchmark and wall metrics.
1200	0.1358901676	0.1537547946	0.0006734136	Best overall result and promoted pass-five winner.

Table 1: Pass-five convergence study for the preserved SST baseline.

The monotonic improvement across all three metrics is the strongest pass-five result. It shows that the pass-four winner was still materially under-converged, and that continuing the baseline was a higher-value action than changing the model or geometry prematurely.

7 Turbulence-Model Comparison

The requested model screen was kept narrow and honest:

- `kOmegaSST` on the official-grid-inspired geometry;
- `SpalartAllmaras` on the same geometry;
- `SA-RC` documented as unavailable in the local OpenFOAM installation.

The SST geometry-fidelity case completed and provided a usable comparison. The `SpalartAllmaras` case initialized and advanced one iteration, which confirmed that the model-specific field generation was valid, but it did not produce a usable converged field in this setup. Because pass five is required to stay reproducible and physically honest, that SA run is documented as a failed screen rather than promoted or massaged into a comparison.

8 Cp And Cf Comparisons Versus Official NASA Data

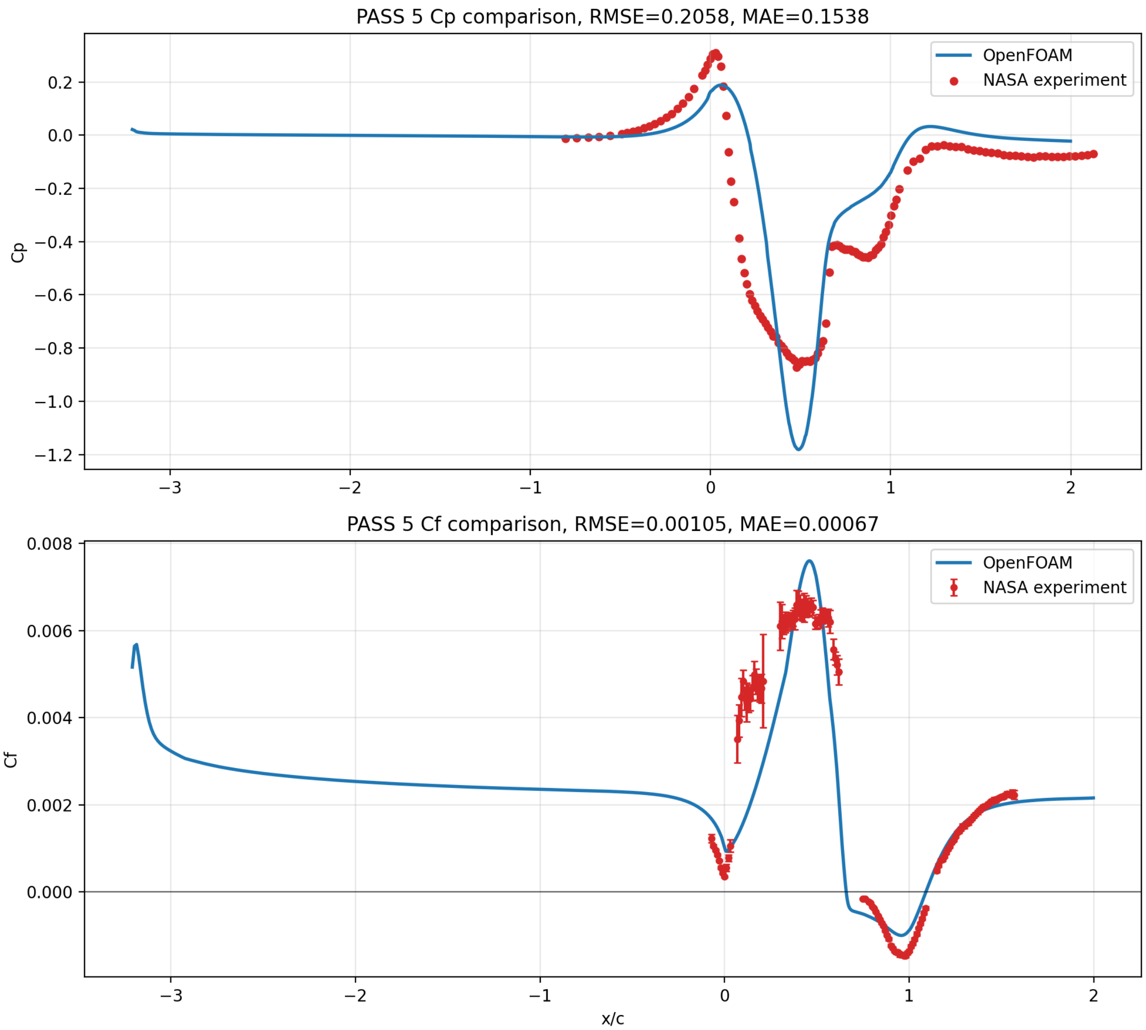

Figure 1: Pass-five comparison against the official NASA no-flow-control C_p and C_f tables using the promoted `1200`-iteration SST continuation.

The promoted pass-five winner reached:

- C_p RMSE: `0.2058487616`
- C_p MAE: `0.1537547946`
- C_f RMSE: `0.0010499684`
- C_f MAE: `0.0006734136`

Compared with the pass-four winner, the pressure agreement improves across the same wall series and the skin-friction mismatch narrows again. The remaining mismatch is still concentrated around the front-half suction behavior and the detailed wall-shear evolution across separation and recovery, but the solution is moving in the correct physical direction without any artificial tuning.

9 MAE Table Across All Pass-Five Experiments

Experiment	Time	MAE	C_p MAE	C_f MAE	Major changes	Diagnosis
Pass-4 base	650	0.1659136747	0.1711800326	0.0008476129	Promoted pass-four SST baseline	Reference checkpoint; still under-converged.
SST cont.	800	0.1547381028	0.1641787966	0.0007681825	Same SST case, longer continuation	Immediate metric improvement with no physics changes.
SST cont.	1000	0.1433377644	0.1578824614	0.0006999053	Same SST case, longer continuation	Continued reduction in wall and benchmark error.
SST cont.	1200	0.1358901676	0.1537547946	0.0006734136	Same SST case, longer continuation	Best overall result and promoted winner.
Grid-fit SST	800	0.1866349502	0.1730924866	0.0008231240	Broader top wall and redistributed streamwise blocks	Slightly better C_f than pass four, but worse benchmark MAE.
Grid-fit SA	failed	–	–	–	Same improved geometry with SA	Did not produce a usable converged field.

Table 2: Pass-five experiment ranking and promotion logic.

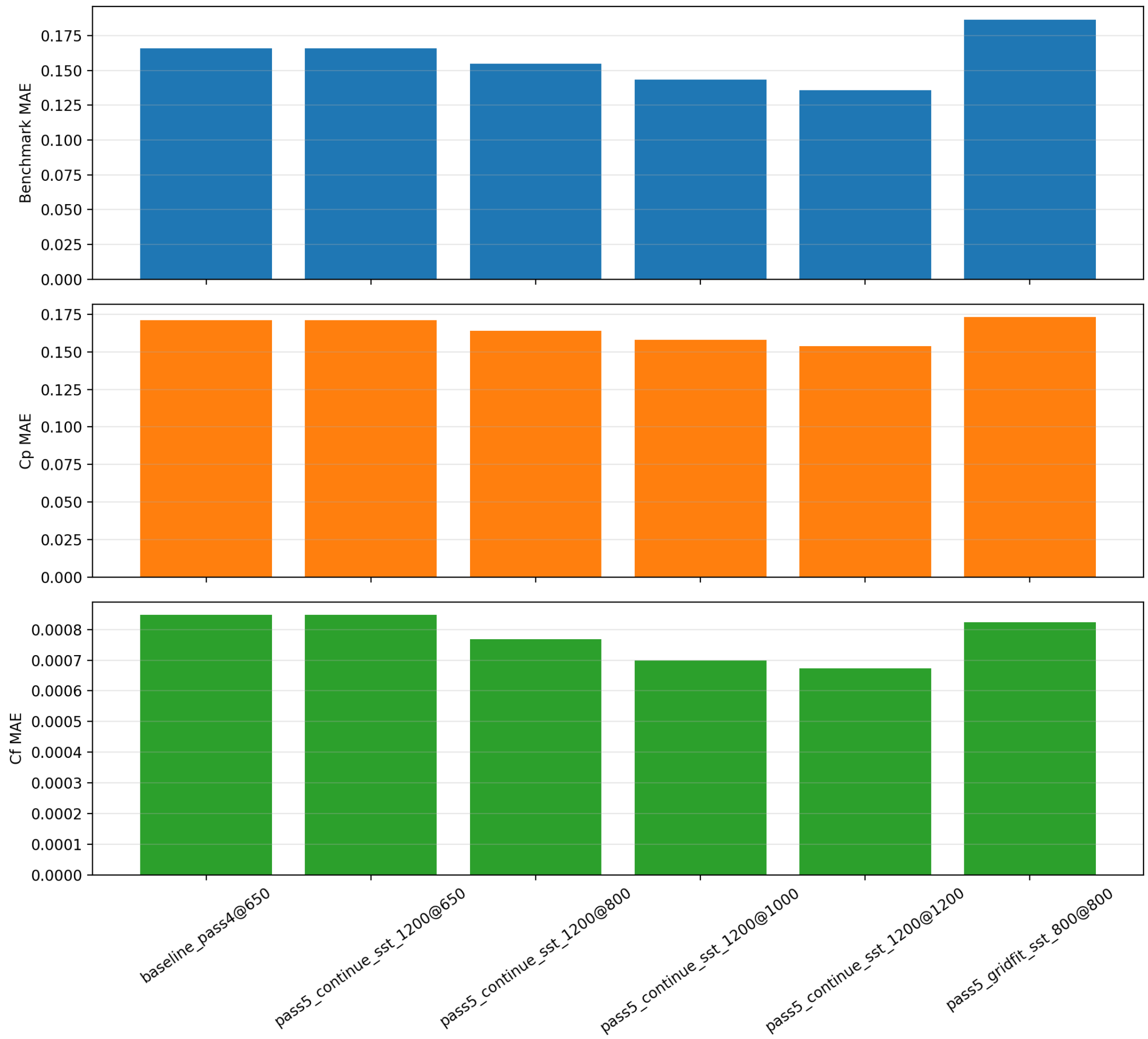

Figure 2: Benchmark MAE, C_p MAE, and C_f MAE across the promoted pass-five experiments.

10 CFD-Derived Graph Set

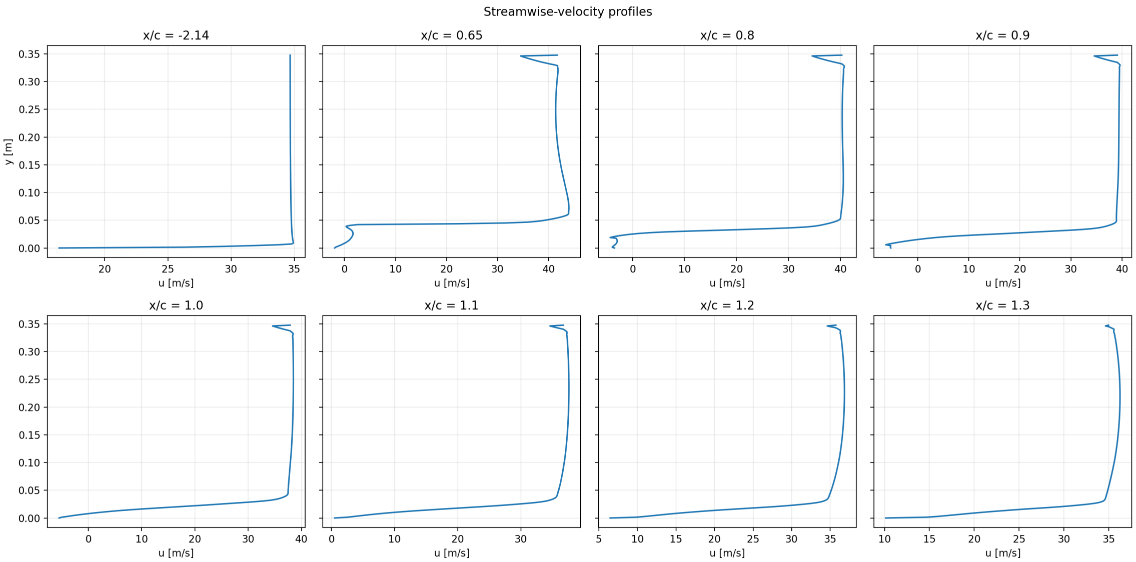

Figure 3: Velocity-profile set regenerated from the promoted 1200-iteration field.

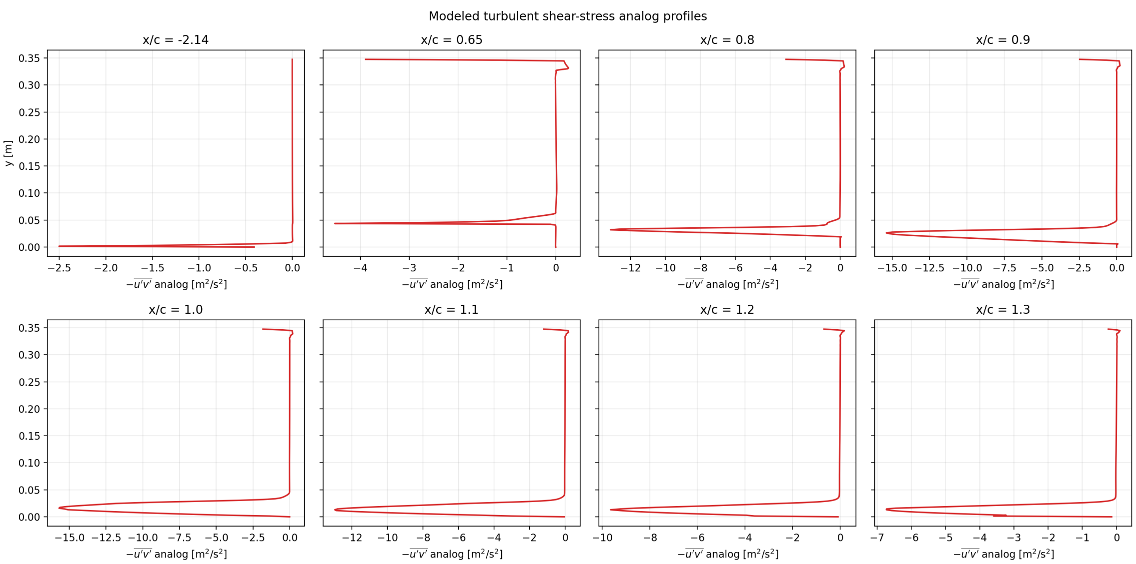

Figure 4: Turbulent-shear analog profiles regenerated from the promoted pass-five field.

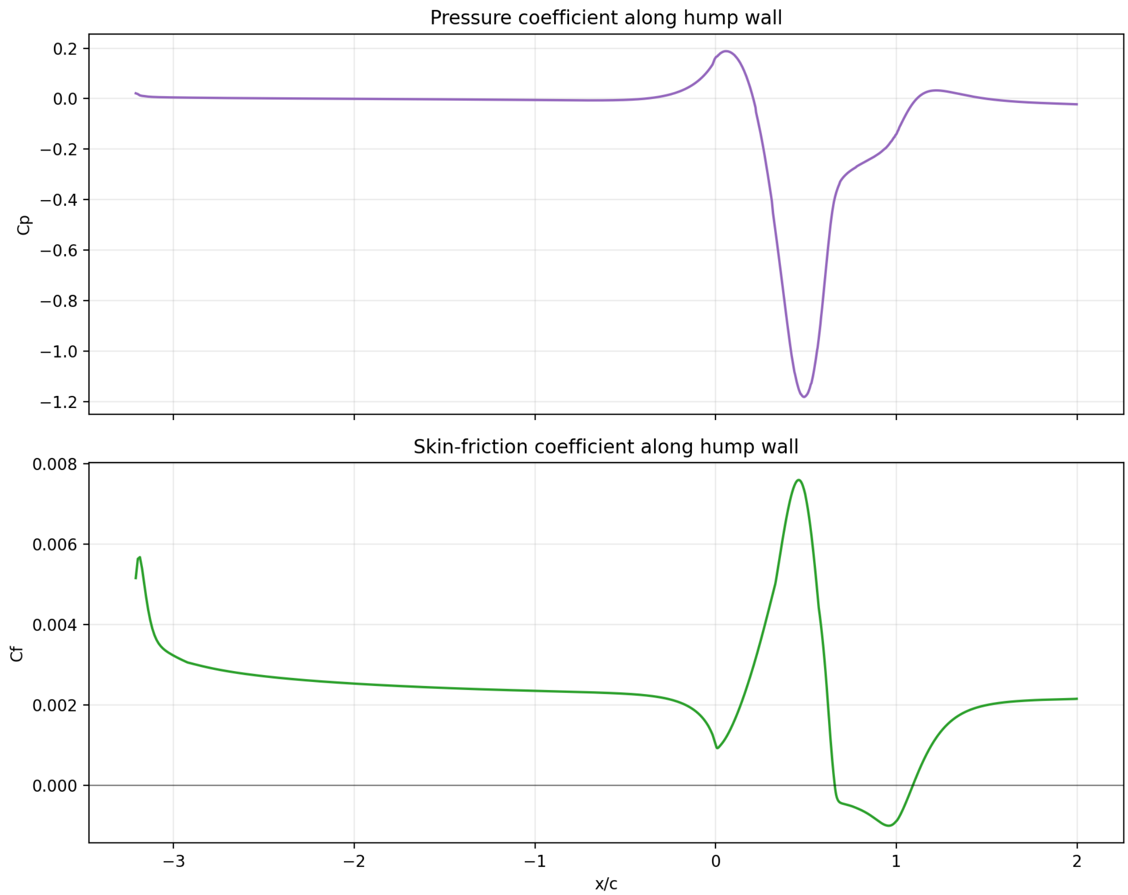

Figure 5: CFD-only wall C_p/C_f distribution derived from the promoted pass-five field.

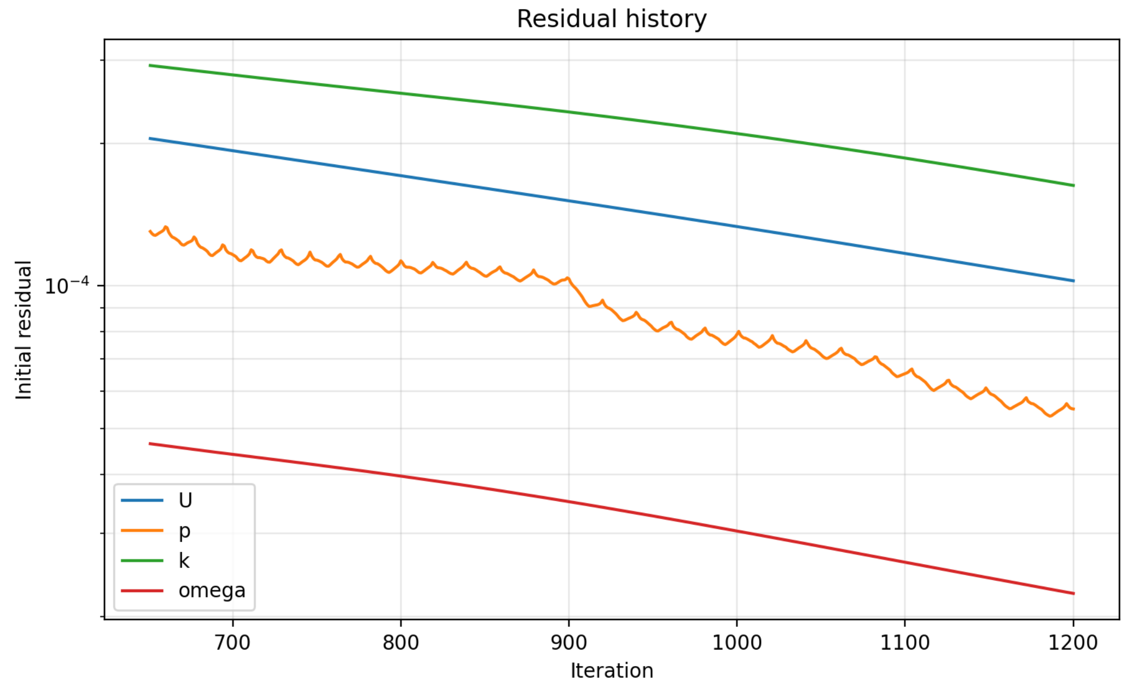

Figure 6: Residual history from the promoted pass-five winner.

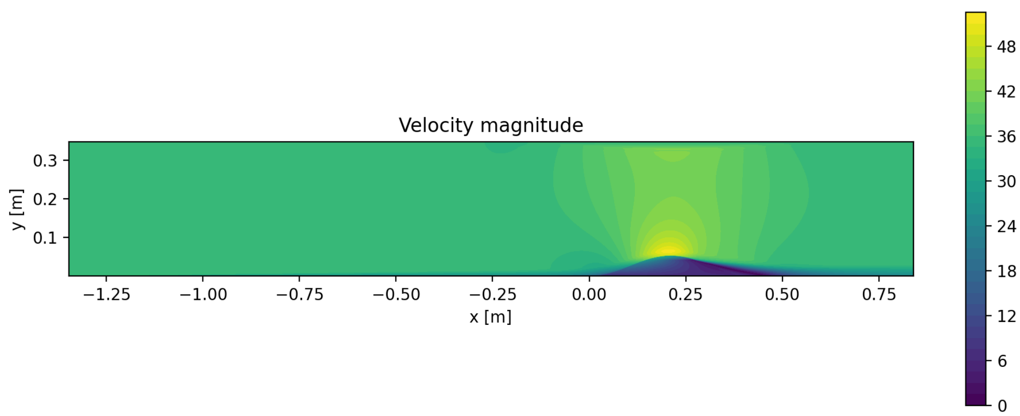

Figure 7: Velocity contour from the promoted pass-five field.

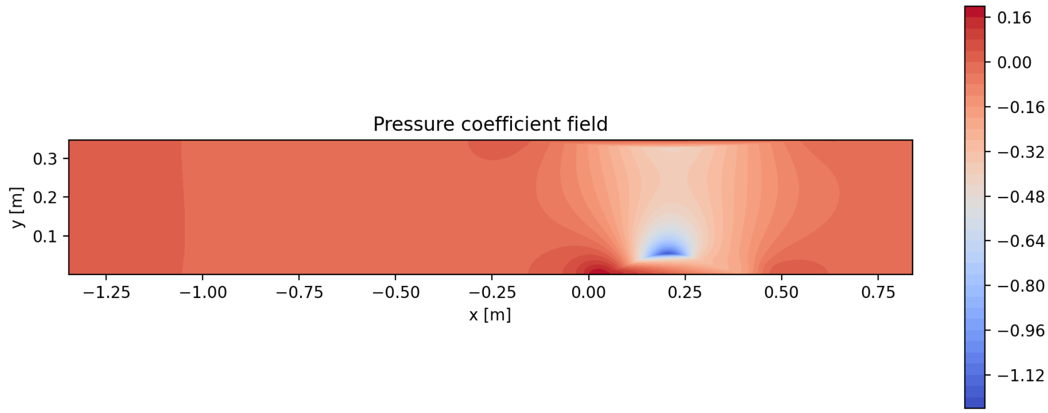

Figure 8: Pressure contour from the promoted pass-five field.

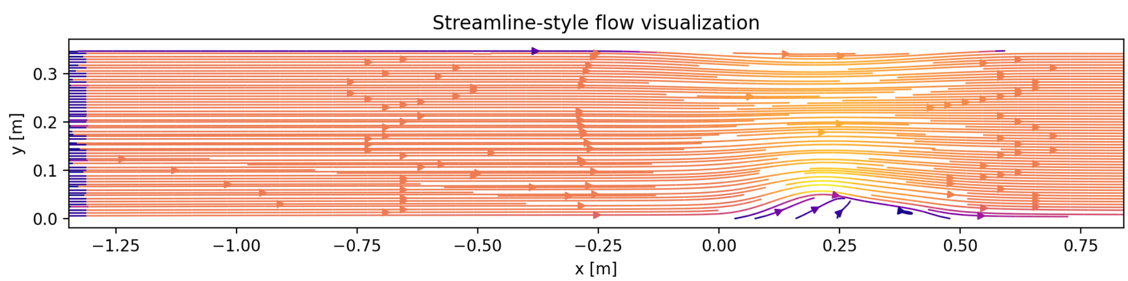

Figure 9: Streamline-style visualization from the promoted pass-five field.

11 ParaView Outputs

The pass-five package keeps the live `.foam`-based ParaView workflow and refreshes the screenshots from the promoted case under `data/NASA_2DWMH/foam.foam`.

Figure 10: ParaView velocity rendering from the promoted pass-five `.foam` case.

Figure 11: ParaView pressure rendering from the promoted pass-five `.foam` case.

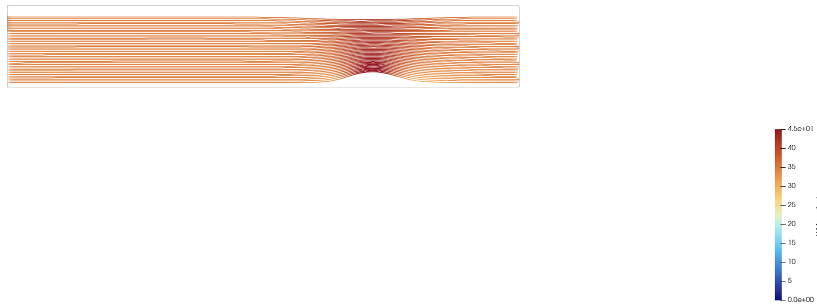

Figure 12: ParaView streamline-style rendering from the promoted pass-five `.foam` case.

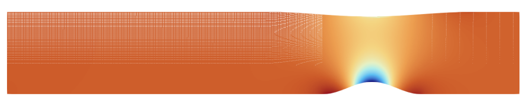

Figure 13: ParaView mesh-wireframe view of the promoted pass-five case.

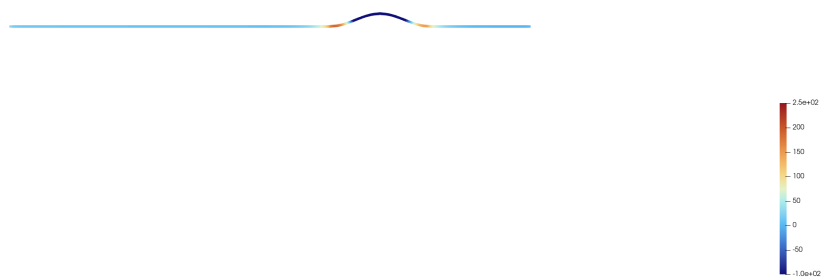

Figure 14: ParaView wall-focused view of the promoted pass-five case.

12 Discussion

Pass five gives a sharper answer than the earlier passes. The most effective remaining improvement was not a new turbulence closure and not a more aggressive geometry change. It was simply to continue the already-best SST case until the residual trajectory and the evaluated metrics flattened much farther downstream than pass four reached.

This matters for two reasons:

- it improves the benchmark MAE and the official wall-data agreement at the same time, which is much easier to defend physically than a one-metric improvement;
- it reduces the temptation to attribute remaining error to model choice before the baseline has been converged as far as practical.

The official-grid-inspired geometry case was still worth running because it tested one of the largest remaining approximations directly. Its outcome was informative: the change did not beat the simpler converged continuation, which means the current metric is still more sensitive to convergence fidelity than to this first geometry-fidelity adjustment.

The SA screen was also useful even though it failed. It confirmed that a model change is not automatically helpful in this reconstructed setup, and that pass five should promote only a case that produces a stable, reproducible solved field.

What still mismatches the official NASA trends:

- the front-half suction peak still does not match the experiment perfectly;
- the detailed wall-shear evolution around pre-separation and recovery still differs from the official C_f trend;
- the geometry is still a reconstruction rather than a direct imported official no-plenum grid.

13 Conclusion

Pass five improved the local best NASA hump benchmark MAE from `0.1659136747` to `0.1358901676`. The winning case was the most conservative one: the promoted pass-four `kOmegaSST` baseline continued from `650` to `1200` iterations.

This is a stronger result than a more complicated model or geometry change because it improves:

- benchmark MAE;
- official C_p agreement;
- official C_f agreement;
- reproducibility;
- numerical stability.

The next best path, if a future pass is attempted, is not a blind sweep. It is a more faithful official-grid-based geometry import or reconstruction that can be paired with the already-proven deeper-convergence workflow and then screened carefully against the current `1200`-iteration SST winner.

14 Reproducibility

From the repository root, the pass-five workflow is:

```
python3 scripts/import_nasa_hump_experimental_data.py
bash scripts/run_nasa_hump_pass5_experiments.sh
python3 scripts/evaluate_nasa_hump_second_pass.py
MPLCONFIGDIR=docs/.mplconfig python3 scripts/make_nasa_hump_second_pass_figures.py
MPLCONFIGDIR=docs/.mplconfig python3 scripts/make_nasa_hump_pass5_figures.py
bash scripts/render_nasa_hump_paraview.sh
bash scripts/build_report.sh
```

References